Medical Education in the Time of Tech Transformation



DR ANURAG AGRAWAL

THE HEALTHCARE LANDSCAPE is evolving rapidly, driven by advancements in technology, changes in population demographics, commoditisation of information in a digital world, and the ongoing need for quality care. While physicians play a critical role in defining and delivering patient care, there is a growing recognition of the importance of non-physicians in imagining and creating health systems of tomorrow. Health is a superficially simple word, hiding a complexity such that its meaning changes from birth to death, from past to present, and from resource-constrained to resource-abundant. The future of health and healthcare systems is being shaped by new scientific, social and economic forces. In turn, medical education and research too will change; with the rise, fall and realignment of specialties. Here, I look at recent technological trends with the potential to disrupt medicine, and speculate about what it means for medical education and research.

The advent of artificial intelligence (AI) and digital health: While the practice of medicine today is not very different from what it was before the turn of the century, the next 25 years are expected to see radical change. Digital health, seen in a wider meaning of the convergence of digital connectivity, wearable devices, digital twins and AI, will embody this transformation. The last two years have seen breathtaking advances, with generative AI coming to the forefront. Whether answering difficult questions, given all relevant information, or holding fully interactive medical conversations with patients to obtain such information, large language models (LLM) have shown the potential for versatile applications. Image classification was already an area where AI may be better than humans, and now we have large multimodal models (LMM) that can handle multiple types of data. While humans still need to be in the



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